JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

DEPARTMENT OF CHEMISTRY

[CHEMISTRY LABARATORY MANUAL FOR FIRST YEARS]

THIS MANUAL PROVIDES STUDENTS WITH BASIC KNOWLEDGE OF HANDLING CHEMICALS AND PERFORMING EXPERIMENTS IN A CHEMISTRY LABORATORY.

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ALL STUDENTS SHOULD THOROUGHLY UNDERSTAND AND ADHERE TO THESE NOTES DURING PRACTICAL SESSIONS.

Introduction

Performing experiments with chemicals in the laboratory is one of the most important and exciting aspects of chemistry. It is from the results of experiments over years that the information presented in lectures has been discovered. The search for further insight into the underlying principles of chemistry, for new compounds, particularly of biological significance, for new uses of compounds, and for information about the secrets of the chemistry of living organisms continues in the laboratory.

However, chemical laboratories can be very dangerous places in which to work.

The following general safety precautions should be observed by every student whenever working in the laboratories:

1. You must assume all chemicals to be toxic unless you are specifically instructed by a member of the staff to the contrary.
2. No food or drink should be consumed in the laboratory.
3. No smoking in the laboratory.
4. You must wear a laboratory coat.
5. No bare feet.
6. Long hair and loose clothing must be confined with rubber bands or safety pins while working in the laboratory.
7. **NEVER** heat flammable liquids, even in small amounts, with a flame, unless the liquid is in a flask with a condenser attached. Do not pour flammable liquids from container to another if a flame is near. Before lighting a burner, check with those working around you to determine if it is safe to do so.
8. **NEVER** heat a closed system of any kind.

9. Keep all chemicals away from your face. Do not measure, heat or mix any chemicals in front of your face.

IMPORTANT TECHNICAL POINTS

1. Keep your working spaces neat at all times and clean up before you leave at the end of the period.
2. When boring a cork do not bore against the palm of your hand.
3. When forcing glass tubing through a cork or stopper, do not use any part of your body as a backstop for the tubing. Hold the tubing as close to the cork or stopper as possible, preferably with a piece of cloth.
4. Use a flexible metal spatula to break up caked solids in bottles, not a glass rod.
5. Use Erlenmer flask for re-crystallization not beakers.
6. Do not place volatile solvents (often indicated in solvents bottles) in an open flask except for a very short period of time.
7. NEVER assemble apparatus over a sink or delivery distillate into a sink.
8. Do not evacuate a flat-bottomed flask unless it is a heavy-wall suction flask
9. Materials which give off noxious fumes should be handled in fume hood.
10. Dispose of organic solvents into the waste recovery bottle.
11. Always wash your hands before leaving the laboratory.

A WORD OF WARNING

Most of the apparatus used are expensive to buy so use them carefully. If you are in doubt whatsoever as to how to assemble or use the apparatus, please consult a Technician or Lecturer in charge. Practical write up must be handed in within the specified time given and not later. Nobody should hand a report for a practical that he/she never attended.

UNIT SCH 2100:

INORGANIC CHEMISTRY ONE.

EXPERIMENT 1.

VOLUMETRIC ANALYSIS.

Introduction

A quantitative analysis based upon the measurement of volume is called volumetric or titrimetric method. Volumetric methods are much more widely used than gravimetric methods because they are usually more rapid and convenient. In addition they are often as accurate.

Procedure

Weigh out accurately about 1.32 g of a substance which is a metal carbonate with the formula X_2CO_3 into a 250 ml volumetric flask. Add about 100 ml of distilled water and stir until the crystals dissolve. Adjust the volume of the solution in the volumetric flask to the mark. Pipette 25 ml of this solution into a 250 ml conical flask. Add 2-3 drops of methyl Red indicator and titrate with a standard 0.1 hydrochloric acid. Repeat the titrations until the titres agree to 0.05 cm^3. Record your results in a table.

Calculations.

(H=1.0 CL=35.5 C=12.0 O=16.0**)**

(a) Write a balanced chemical equation for the reaction between hydrochloric acid and X_2CO_3 carbonate solution.

(b) (i) How many of the acid took part in the reaction?

(ii) Hence calculate the molarity of the carbonate solution in moles/dm^3

(iii) Also calculate the concentration of the metal carbonate solution in g/dm^3.

(c) Calculate the relative formula mass (R.F.M) of the metal carbonate X_2CO_3.

d) Calculate the relative atomic mass (R.A.M) of X.

e) Identify metal X with the help of a periodic table.

EXPERIMENT 2.

Visual observations of Emission colors of some of the Alkali metals

Theory

When the alkali metals are heated their outermost electrons are easily excited to higher energy states. When these excited electrons 'drop back' to the ground energy states, each alkali metal emits a characteristic color (which occurs in the visible region hence a visual observation).

Procedure

Make appropriate dilute solutions of the salts $NaCl_2$, KCl, LiCl and use distilled water to make the above solutions. Dip a platinum wire in each solution and quickly remove it and put it on the flame. Note the color each sample produces. Repeat the process in a tap water. Repeat the process in a solution $CaCl_2$. In your write up, identify the most dominant alkali in the tap water.

Exercises:

a) Draw an energy level diagram (sketch) which roughly explains how the above colors are produced. Explain the process involved.
b) Draw a table showing the colors emitted by the elements Lithium, Sodium, potassium, ceasium, and Calcium.
c) Explain the major difference between the colors produced by the alkali metals and Calcium.

EXPERIMENT 3:

STANDARDIZATION OF HCL SOLUTION (NON PRIMARY STANDARD) SOLUTION CARBONATE AS A PRIMARY STANDARD.

Introduction

The process by which the concentration of a chemical species is determined is known as standardization. A primary standard solution is one whose concentration is known. In this case the type of reaction used is that of ACID-BASE TITRATION

Reaction: $Na2CO3+2HCl \longrightarrow 2NaCl+CO_2+H_2O$

Procedure

Weigh out accurately about 1.3 g of primary standard sodium carbonate into a 250 ml volumetric flask, add about 100 ml of distilled water and shake until dissolved. Adjust the volume to the mark and mix thoroughly. Pipette 25 ml of this solution into a 250 ml conical flask, add 2-3 drops of methyl red and titrate with HCl solution to be standardized until the solution turns brown red. Now boil the solution for 30 seconds. The color of the solution should return to yellow. Cool the solution and titrate until the red appears again. Boil the solution and if the yellow color returns again, repeat the above procedure. The titration is complete when the red color persists.

Calculations:

Repeat until titres agree to 0.05 ml.

1. Calculate the molarity of the Na_2CO_3 solution.
2. Give the volume of the HCl used.
3. Calculate the molarity of the HCl used.
4. Calculate the concentration of HCl in g/l
5. What is the equivalent weight of Na_2CO_3?
6. What is the concentration of the HCl in normality units, in the above reaction?

Reference book: Quantitative inorganic analysis by A.L Vogel.

EXPERIMENT *4*:

STANDARDIZATION OF APPROXIMATELY 0.1 M SODIUM HYDROXIDE USING AN ORGANIC ACID AS A PRIMARY STANDARD.

Introduction:

Analytical Reagent (A.R) potassium hydrogen, phthalate has a purity of at least 99.9%. It is almost non-hygroscopic, but unless a product of guaranteed purity is purchased, it is advisable to dry it at 120 ^{0}C for two hours and allow it to cool in a covered vessel in desiccators. With a carbonate free sodium hydroxide, titration using phenolphthalein or thymol blue as the indicator may be employed.

Reaction: HK ($C_3H_4O_4$) +H_2O

Procedure:

Weigh out accurately 2.04 g of the ordinary Analar (A.R) product of potassium hydrogen phthalate into a 100 ml volumetric flask. Add distilled water and dissolve the solid. Make up the solution to 100 ml (up the mark). Using 10 ml portions of this solution titrate each of them with the sodium hydroxide solution (approximately 0.1M already prepared) contained in a burette, using the phenolphthalein or thymol blue indicator.

Note: Individual titrations should not differ by more than 0.1 ml.

Questions:

1. Calculate the concentration of sodium hydroxide using both molarity and normality unit system.
2. a) What is the equivalent weight of potassium hydrogen phthalate?
   (b) How many gram equivalents are there in the2.04 g of the salt?
   (c) What was the concentration of the salt in both normality and molarity unit systems?

Reference:

Quantitative inorganic analysis, by A.L Vogel

EXPERIMENT 5

REDOX REACTIONS USING POTASSIUM PERMANGANATE

Introduction

Potassium permanganate is probable the most widely used of all volumetric oxidizing agents. It is a powerful oxidizing agent and readily available at moderate cost. The intense color of the permanganate ion is sufficient to signal the end point in most titrations, this eliminates the need for an indicator.

The tendency of permanganate to oxidize chloride ions represents a limitation since hydrochloric acid is often a desirable solvent. Furthermore, solutions of permanganate have limited stability.

In this experiment you will titrate a standardized 0.2 M $KmnO_{4\,(aq)}$ solution with aqueous solutions of iron (II), H2O and oxalic acid. The half reactions involved in the reactions are given below:

$MnO_4 + 8H + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$Fe^{2+} \longrightarrow Fe^{3+} + e^+$

$H_2O \longrightarrow O_2 + 2H^+ + e^-$

$C_2O_2^{2+} \longrightarrow 2CO_2 + 2e^-$

OXIDATION REACTIONS OF PERMANGANATE ION.

a) **Oxalic acid**

In acid solution, permanganate oxidizes oxalic acid to carbon dioxide with water. This reaction is slow at room temperature, but above 600 C it is fast enough to be useful in analysis. The reaction catalyzed by the manganese (II) ion, which is formed during the titration, autocatalysis.

b) **Hydrogen peroxide**

Hydrogen peroxide is oxidized to oxygen by the permanganate ion in acid solution; the reaction is fast at room temperature unlike the reaction

in (a). Note that very dilute solutions of hydrogen peroxide decompose rather quickly.

c) **Iron(II)**

Perhaps the most important permanganate method is the analysis for iron ores, steels and other alloys. Iron (III) in acid solution.

TITRATION OF AN OXALIC ACID WITH STANDARD POTASSIUM PERMANGANATE

Pipette 20 ml of oxalic acid solution into a 250 ml conical flask. Add about the same volume of 1M of sulphuric acid and heat to about 80^0 C. Titrate to a faint color, indicating a small excess of permanganate. Repeat the procedures until three titrations differ by a maximum of 0.2 ml. Record your results in a table.
Calculate the concentration of the oxalic acid solution.

TITRATION OF HYDROGEN PEROXIDE

Pipette 20 ml of 20-volume hydrogen peroxide solution. Acidify with about the same volume of 1M sulphuric acid and titrate with the standardized permanganate solution.
Repeat titrations at least twice. Calculate the concentration in molarity and percentage by weight (%W), assuming that the density of peroxide solution is 1.0 g/ml.

TITRATION OF IRON (II) SOLUTION

Mohr's salt, $(NH_4)2Fe\ (SO_4)$, $6H_2O$, keeps well under storage (it does not lose water or become air-oxidized). Weigh accurately 1 to 1.5g of the

salt into a 250 ml conical flask. Dissolve the salt in about 40 ml, 1M sulphuric acid and titrate with standardized permanganate solution. Repeat at least once. Calculate the percentage of iron in the salt and compare your results with the theoretical, two more titrations.

References

1. A.I. Vogel, a textbook of Quantitative Inorganic Analysis, Longman, London(1971)
2. L.B. Clapp, Investigating Chemical Systems II, Rhode Island(1971)

Exercises

1. Write net ionic equations (redox equations) for the three reactions in part (a to c).
2. Calculate the concentration of iron(II) solutions and that of oxalic acid in:
   i) moles/litre
   ii) grams/litre
3. Give two reagents you can use to reduce iron (III) to iron (II).
4. Explain briefly the use of redox titrations as an analytical method.
5. Write the equations for the oxidation reactions for potassium permanganate in alkaline media.
6. Why is potassium permanganate not used as a primary standard?

EXPERIMENT 6:

TITRATION OF ANTACIDS

Introduction

The purpose of this experiment is to determine the amount of solid neutralized by various commercial antacids. These products contain bases such as calcium carbonate, magnesium carbonate and magnesium hydroxide. The latter compounds are not very soluble in water, but direct titration can be carried out with hydrochloric acid if sufficient time is allowed for the reaction the solid and the titrant. A recurring and point may be obtained because this reaction is rather slow.

In the procedure below, excess acid is added to react with the antacid, the solution is heated to remove CO2, and the excess acid is titrated with standard base. Phenolphthalein can be used as an indicator and reasonably sharp end point is obtained.

$Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + H_2O$

$MgCO_3 + 2HCl \longrightarrow MgCl_2 + CO_2 + H_2O$

Procedure:

1. Take half a tablet of antacid and weigh on the analytical balance.
2. Transfer the sample of a 250 ml Erlenmeyer flask.
3. Add 50 ml of standard 0.1M HCl.
4. Heat the solution to boiling then boil it gently for about 3 minutes.

5. Cool the solution to room temperature.
6. Add 4 drops of the indicator and titrate with standard base to the first permanent pink color.
7. Calculate the grams of HCl neutralized by 1 g of antacid.
8. Assume that 0.1M HCl has density of 1.00g/ml
9. Calculate the grams of 0.1m HCl solution neutralized by 1g of the antacid.

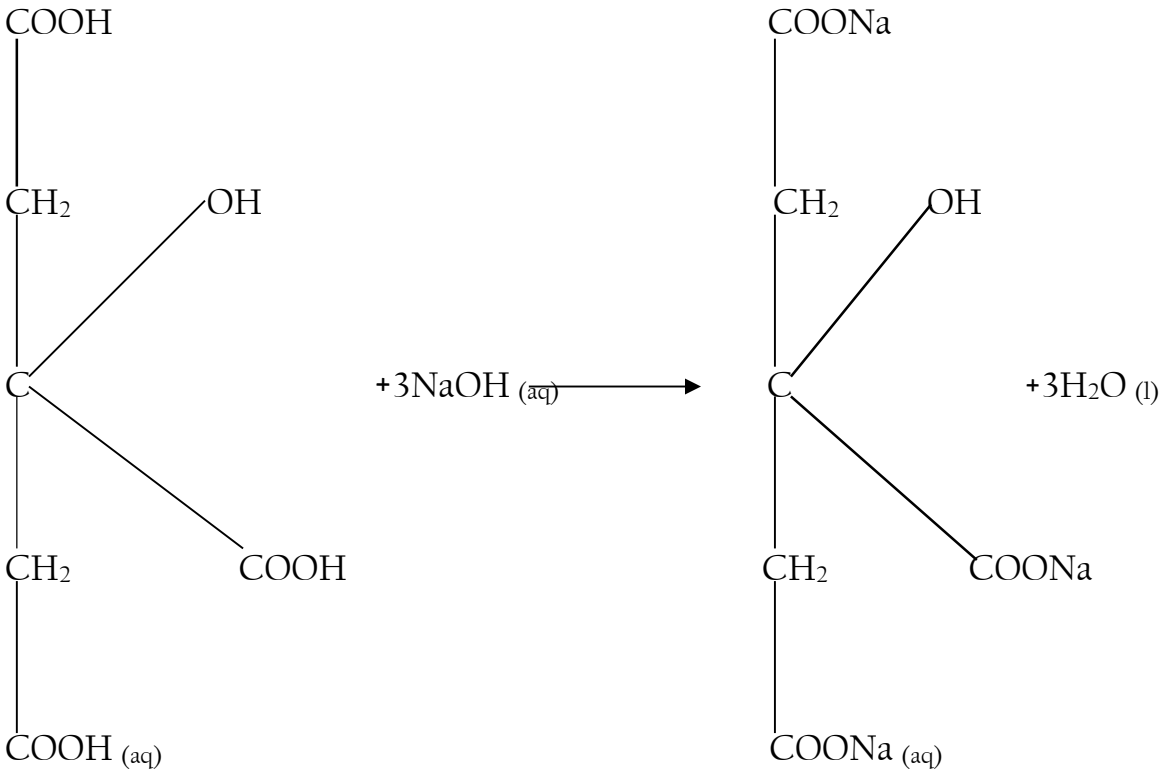
Note:

There may be a small amount of white solid (filler), which does not dissolve even after heating.

EXPERIMENT 7:

DETERMINATION OF CITRIC ACID IN ORANGE SQUASH/FRUIT JUICES.

You are provided with orange squash or commercial fruit juice that has been diluted 50%. It is alleged to contain citric acid has indicated in the assay. The Government National Laboratory wants to ascertain the concentration of citric acid in the commercial juices. The aim of this experiment is to find the concentration of the citric acid given that it reacts with NaOH by the equation:



Procedure.

Put the squash in the burette. Take 10ml aliquots of the 0.1M NaOH provided into a conical flask. Add phenolphthalein indicator and titrate this against the squash.

Calculations

1. Calculate the number of moles of NaOH in 10cm^3 of the 0.1M NaOH
2. How many moles of citric do these moles of NaOH react with?
3. What is the volume of citric acid containing these moles?
4. What is the original undiluted volume of the squash that contained these moles of citric acid?
5. Calculate the concentration of the original acid in moles per litre.
6. Calculate the concentration of the citric acid in grams per litre (C=12, O=16, H=1).

EXPERIMENT 8:

DETERMINATION OF ACETIC ACID CONTENT OF VINEGAR

The principal acid of vinegar is acetic acid, and federal standards require at least 4g of acetic acid per 100ml of vinegar. The total quantity of acid can be readily determined by titration with standard base using phenolphthalein indicators. Although other acids are present, the result is calculated as acetic acid.

Procedure.

Pipette 25ml of vinegar into 250ml volumetric flask, dilute to the mark and mix thoroughly. Pipette a 50ml aliquot of this solution into an Erlenmeyer flask and add 50ml of water and 2 drops of phenolphthalein indicator.
Titrate with the standard base to the first permanent pink color.
Repeat the titration on two additional aliquots.
Assuming all the acid to be acetic, calculate the number of grams of acid per 100ml of vinegar solution. Assuming that the density of vinegar is 1.00, what is the percentage of acetic acid by weight in vinegar? Average your results in the usual manner.

EXPERIMENT 9

To find the equation for reaction by titration

Put potassium permanganate (0.02M) in the burette. Pipette 25cm^3 of oxalic acid solution into a conical flask. (CARE: oxalates are poisonous) and add 10cm^3 of 1M sulphuric acid. Heat the mixture to about 60 ^{0}C and then titrate against the permanganate solution until a permanent pink color is obtained.

Questions

1. Find the number of moles of permanganate in the mean titre and hence calculate the number of moles of oxalic acid that react with the permanganate in the 25 cm^3 of oxalic acid solution.
2. How many moles of oxalic acid react with one mole of permanganate?
3. How many moles of oxalic acid react with two moles of potassium permanganate?
4. The half equation are:

   $MnO_4^-{}_{(aq)} + 8H^+ + 5e^- \longrightarrow Mn^{2+}{}_{(aq)} + 4H_2O_{(l)}$

   $(COO)^{2-}{}_{2\,(aq)} \longrightarrow 2CO_{2\,(g)} + 2e^-$

   Electrons given by the oxalate ion $(COO)^{2-}$ are taken up by MnO_4^- and H^+. Write down an overall balanced equation for the reaction.
5. Assigning the manganese ion MnO_4^- n positive changes, calculate n.

EXPERIMENT 10:

REDOX TITRATION USING POTASSIUM DICHROMATE AS A PRIMARY STANDARD

Introduction:

The term oxidation was originally applied to reactions involving the reaction of oxygen with another element or with a compound. Likewise the term reduction was used to indicate removal of oxygen from a compound.

Oxidation in the broad definition refers to loss of electrons and reduction to gain of electrons. A substance that undergoes oxidation brings about the reduction of another species. It is therefore a reducing agent; the substance responsible for oxidizing another substance is called an oxidizing agent. In a redox reaction therefore the loss of electrons by one species is accompanied by the gain of electrons by another species.

Potassium dichromate, $K_2Cr_2O_7$, is a good primary standard for redox titrations; it is obtainable in pure, can be dried without decomposition, has a relatively high molecular weight, and dissolves readily to give stable solutions.

In acidic solutions $K_2Cr_2O_7$ reacts quantitatively according to the equation:

$$K_2Cr_2O_7 + 14H^+ + 6e^- \longrightarrow 2K^+ + 2Cr^{3+} + 7H_2D$$

I.e. $$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow aCr^{3+} + 7H_2O$$

Since there is gain of electrons on the left hand side of the equation, the dichromate ion, CrO^{2-}_7 is reduced. The corresponding species in the titration will have to undergo oxidation to supply the electrons that are taken up by the dichromate ion.
In this experiment iron (II) will be as the reducing agent, iron (II) is oxidized to iron (III) by a suitable oxidizing agent, such as potassium dichromate, according to the equation:

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

Since one mole of dichromate ion requires six electrons, then six moles of iron (II) will be required for every mole of dichromate ion. i.e.

$$6Fe^{2+} \longrightarrow 6Fe^{3+} + 6e^-$$

The overall balanced redox reaction is:

$$6Fe^{2+} + Cr_2O^{2-}_7 + 14H^+ \longrightarrow Fe^{3+} + 2Cr^{3+} + 7H_2O$$

The end point of dichromate titration is detected by using a suitable redox indicator; a color change from deep green to intense violet-blue occurs with barium dephanylamine sulphate.

Requirements

— 2 burettes and one 1 litre volumetric flask.
— 50 or 100cm^3 beakers
— A.R potassium dichromate solid
— Technical grade ferrous ammonium sulphate
— 0.3% aqueous diphenylamine indicator
— Conc. 90% orthophosphoric acid.

Procedure

Preparation of standard $K_2Cr_2O_7$:

Weigh accurately 50.8g (to the nearest 0.001g) of dried $K_2Cr_2O_7$ and empty the contents into 1 litre volumetric flask. Add sufficient distilled water and mix well to dissolve the $K_2Cr_2O_7$. Add enough water to the 1 litre mark ().

Preparation of standard solution of Ferrous ammonium sulphate

Iron (II) is rather unstable in air and consequently the samples must **not** be heated in oven to dry them.

Weigh accurately 390.g (the nearest 0.001g) of ferrous ammonium sulphate and empty the contents into a 1 litre volumetric flask. Add sulphuric acid to dissolve the solid and bring to 1 litre mark with distilled water. Use of sulphuric acid as a solvent prevents hydrolysis and also provides the necessary acidic conditions for the titrations.

Fill the burette with $K_2Cr_2O_7$ solution. Pipette exactly 25ml of iron ammonium sulphate solution into a 50 or 100 ml flask. Add about 0.5 ml of the indicator solution and about 2ml of conc. Phosphoric acid which reacts with the iron (III) ions, producing a complex which does not affect the indicator.

Carry out one rough titration and three accurate titrations. Results should agree to within $0.05cm^3$.

Results and Calculations

1. From the exact weight of $K_2Cr_2O_7$ used, calculate the exact concentration (mol^{-1} or M) of your solution.
2. From the titration results, calculate the concentration (mol^{-1}) of your iron ammonium sulphate solution and hence the grams per litre as determined volumetrically.

UNIT SCH 2101.

CHEMICAL BONDING AND ATOMIC STRUCTURE

EXPERIMENT 1:

ACID BASE TITRATION

DETERMINATION OF THE CONCENTRATION OIF SODIUM HYDROXIDE

According to the Brousted-Lowry theory, an acid is defined as a proton donor and a base as a proton acceptor. This is formulated for hydrochloric acid (strong)

$HCl \longrightarrow H^+ + Cl^-$

And for acetic acid (weak)

$CH_3COOH \longrightarrow CH_3OO^- + H^+$

Considering the addition of acetic acid to water, the acid reacts with water according to the following reaction

$HAC + H_2O \longrightarrow H_3O^+ + Ac^-$ where Ac stands for CH_3COO^-(Acetate ion)

All acids behave in an exactly similar way except that with strong acids in fairly dilute solution the hydrolysis is effectively complete, whereas with acids a balanced action is set up (equilibrium). Going by the above equation an-definition, ac' is regarded as a conjugate base of the acid HAC water reacts as a base of the base water.

The above equation can be generalized as:

$Acid_1 + base_2 \longrightarrow Acid_2 + base_1$

Now consider sodium acetate when it is dissolved in water the Ac^- is hydrolysed:

$Ac^- + H_2O \longrightarrow HAC + OH^-$

The base Ac^- accepts a proton from the water to form a **conjugate acid** HAC and a very strong base CH^- is liberated. Water in this case functions as an acid. OH is a **conjugate base** of acid water.

This is one of the many unique properties of water as a solvent. It can react as an indicator base and it is called Ampholyte (short for Amphoteric Electrolyte).

Principles:

$2NaCH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

$2NaOH + H_2SO_4$

Eq.wt. of NaOH = molar mass =40g

Eq.wt of H_2SO_4 = Molar mass =98g/2=49g

0.1N H_2SO_4 = 0.05M H_2SO_4

Reagents:

1. Standard solution 0.1M H_2SO_4 (0.05) containing 4.9g per litre.
2. Alkali of unknown concentration: Sodium hydroxide (NaOH)
3. Methyl red indicator.

Apparatus

1. Pipette(10ml)
2. Burette(50ml)
3. Conical flask(250ml)
4. Small beakers(50ml)
5. White tiles
6. Burette stands.

Procedure:

1. Fill a clean burette with a standard solution of H_2SO_4
2. Pipette in triplicate 10ml of an alkali from a 50ml beaker into a clean conical flask.
3. Add 2 drops of methyl red to an alkali.
4. Then run in the standard solution from a burette until the color changes from yellow to red.

Calculate:

a) 1. Molarity of the alkali

2. Normality of the alkali

b). 1. Grams per litre of the alkali

2. Determine the purity of sodium hydroxide if 4g was dissolved in a litre of solution.

Note:

$N_a \times V_a = N_b \times V_b$ (applicable for normality calculation only).

DATA SHEET

Name:.....Stream.....

Titration

	1	2	3	Average
.....				
ml of H_2SO_4				
.....				

Calculation of results:

$$\text{Normality of NaOH} = \frac{\text{Normality of } H_2SO_4 \times \text{ml. of } H_2SO_4}{\text{ml. of NaOH}}$$

wt. of NaOH per litre= Normality of NaOH x Eq.wt. of NaOH

Calculate:

a). 1. Molarity of the alkali

2. Normality of the alkali

b). grams per litre of the alkali

2. Determine the purity of sodium hydroxide if 4g was dissolved in a litre of solution.

Note:

$N_a \times V_a = N_b \times V_b$ (applicable for normality calculations only).

DATA SHEET

Name:.....Stream:.....

Titration 1 2 3 Average

.....

Ml of H_2SO_4

.....

Calculation of results:

$$\text{Normality of NaOH} = \frac{\text{Normality of } H_2SO_4 \times \text{ml. of } H_2SO_4}{\text{ml. of NaOH}}$$

Wt of NaOH per litre=Normality of NaOH x Eq.wt. of NaOH

$$\text{Purity of NaOH} = \frac{\text{wt. of NaOH per litre} \times 100}{\text{wt of NaOH dissolved in a litre (actual wt)}}$$

$$\text{Molarity of NaOH} = \frac{\text{wt of NaOH per litre}}{\text{Molar mass (molecular weight)}}$$

Report

1. Calculation of normality of NaOH
2. Calculation of wt. of NaOH per litre

3. Calculation of Molarity of NaOH
4. Calculation of purity of NaOH

EXPERIMENT 2:

Acid-Base titration: The titration of weak acid with a strong base

Introduction:

This experiment involves the titration of a weak acid with a strong and the selection of a suitable indicator for the titration. Acid-base titration forms one

branch of volumetric analysis. Other branches such as redox, precipitation, and complex metrics, will be studied later.

The Theory of Acid-base Titrations

Several tutorials have been or will be given on aqueous solutions containing acids and bases and the calculation oh pH of these solutions. At the end of this exercise you will find a problem which should test your understanding of these principles. We will proceed on the basis that you have some understanding.

The purpose of titration, say an alkaline solution with a standard solution of an acid is the determination of the exact amount of acid which is chemically equivalent to the amount of base present. It is not really necessary to say that having this exact amount of acid-we easily calculate the amount of base. The point at which chemically equivalent amounts of acid and base are present is known as the equivalence point or the end point. The ph of a solution at the equivalence point depends on the nature of the acid and the base being titrated. If both are strong electrolytes, the solution will be neutral i.e. $(H^+) = (OH^-)$, and will have a pH of 7. If you titrate a weak acid with a strong base, there will be hydrolysis of the amount of the acid.

$A^- + H_2O = HA + OH^-$

and the solution will have a "basic" pH i.e. >7. Similarly for the titration of a strong acid with a weak base-hydrolysis occurs.

$M^+ + H_2O = MOH + H$ i.e. $MOH + H^-$

And the solution will have a pH >7. Titrations of weak acids with weak bases are usually avoided. The reason for this will become apparent later.

The end point of an acid base titration will therefore be characterized by a definite pH, the value depending on the nature of the acid and the base and as has been discussed in tutorials, the concentration of the solution.

The most important method of detecting the end point of an acid-base titration involves the use of pH indicators. These are substance, either weak organic acids or bases, which posses different colors according to the pH of the solution. Let us

consider the example of a weak acid type indicator Hin. In aqueous solution the acid dissociates according to the equilibrium,

$HIn + H^- + In^-$

First note that the equilibrium is a function of pH. Now if Hin has a different color from that of HIn, the color of a solution containing the indicator will change.

The equilibrium constant for the dissociation is:

$$K_2 = \frac{(H^-)(In^-)}{(HIn)} \qquad (1)$$

Rearranging we obtain $(H^+) + k_a = \frac{(HIn)}{(In^-)}$ (2)

On taking logs for both sides of (2) we have

$$\text{Log } (H^+ = \log K_2 + \log K_a + \log = \frac{(HIn)}{In^-)}$$

Or

$$pH = PH_a + \log \frac{(In^-)}{(HIn)}.$$

In general, the eye will perceive the color of the molecule HIn when it is present to the extent of ten times or more the amount of I^-. Conversely the color of In^- will be perceived when it is present to the extent to ten times or more the amount of HIn.

I.e. In^- color perceived at

10

$pH = pK_a + \log \frac{-}{-} \quad pK_a + 1$

1

HIn color perceived at

1

$pH = pK_a + \log \frac{-}{=} \quad pK_a - 1$

10

This means that the color will change from that of HIn to that of In^- as the pH changes from $pK_a + 1$. Let us take as an example of an indicato0r which has dissociation constant of 10^{-8} i.e. a ph 8. This indicator will change from the color of the molecule HIn to that of the anion In as the pH changes from 7 to 9. Thos interval in pH is known as the pH range of the indicator.

It has been assumed in the above argument that both colors have dual intensity. If this is not true or if the eye is more sensitive to one color than the other there will be some displacement of the pH range.

Let us say that we wish to detect the end point which occurs at pH 7.8. We therefore need an indicator which changes color in the vicinity of this pH. The indicator mentioned above, with a pH range of 7-9, would appear to be likely candidate for this task. Since we process indicators which cover practically every ph range from pH 0 to13, we can usually match an indicator to a titration. However, some very important points need to be stated. The middle tint the color of the solution (when the solution contains equivalent amounts of acid and base) of the indicator applied strictly only if the two colors that of the solution in the acidic range and (that of the solution in the alkaline side are of equal intensity).

If one form is more intensely colored than the other or if the eye is more sensitive to one color than the other, then the middle tint will be highly displaced along the pH range of the indicator.

When doing a titration of a weak acid against a strong base, the end point cannot be accurate since at the equivalence point, the solution contains excess of OH^- ions which will repress the hydrolysis of the salt.

As a general rule it may be stated that for a titration to be feasible, there should be a change of approximately two units of ph at or near stoichiometric point produced by the addition of a small volume of the reagent. The ph at the equivalence point during titration of a weak acid and a strong base is calculated from the equation:

PH=1/2 PKw + ½ pK_a + ½ log C

Where Kw=ionic product of water i.e.

Kw= (H^-) x (OH^-)

K_a=dissociation constant of a weak acid, HA

I.e. K_a= (H^+) x (A^-) (HA)

C=the concentration of the salt

i.e. C=in g-moles per litre

NOTE: The pH range for acids with Ka 10—3 is 7—105; for weaker acids (K 10-5) the range is reduced (8— 10). The pH range 8 10.5 will cover most of the examples likely to be encountered; this permits the use of thymol blue, thymolphthalein, or phenolphthalein.

The weak acid to be used in this experiment is glacial acetic against sodium hydroxide being a strong base.

Objective (a):

(a) Determination of the strength of glacial acetic.

Procedure:

Weigh a dry stoppered 50ml volumetric flask; introduce about 2 ml of glacial acetic acid and weigh again. Add about 20ml of water and transfer the solution

quantitatively to a 250ml graduated flask. Wash the small flask several with water and add the washings to the volumetric flask. Make up to the mark with distilled water. Shake the flask well to ensure thorough mixing. Titrate 10ml portions for the acid with 0.1 M standard sodium hydroxide solution using phenolphthalein or thymol blue as the indicator.

Reaction:

$NaOH + CH_3COOH = CH_3OONa + H_2O$

NOTE: 1 ml of 1M NaOH = 0.06005g. CH_3COOH

QUESTIONS

Calculate the percentage of CH_3COOH in the sample of glacial acetic acid.

EXPERIMENT 3:

DETERMINATION OF THE CONCENTRATION OF HYDROCHLORIC ACID PRESENT IN A GIVEN SOLUTION

The aims of the experiment.

(i) To be able to standardize NaOH solution using a standard solution of Oxalic acid.
(ii) To be able to prepare standard solutions.
(iii) To determine the strength of the given hydrochloric acid solution.

REAGENTS REQUIRED

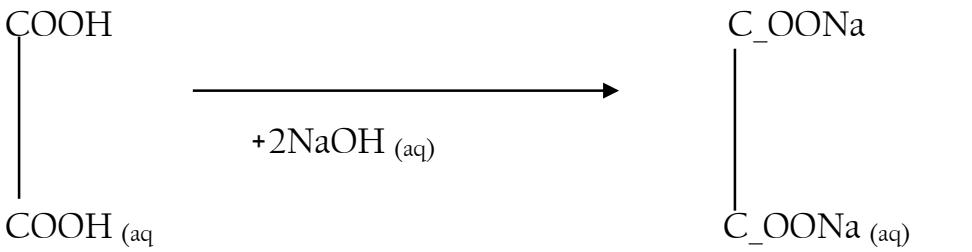
a) Oxalic acid

b) NaOH solution
c) Hydrochloric acid solution of unknown strength
d) Phenolphthalein as an indicator.

Introduction:

A standard solution of sodium hydroxide(NaOH) cannot be made by direct weighing due to its water absorption nature, so a standard solution of some stable acid has to be prepared to standardize the given sodium solution. It is recommended to prepare a standard solution of Oxalic acid by weighing the required amount (0.63g in 250ml) in a weighing bottle by the difference method.

The chemical reactions involved are represented as:



$HCl_{(aq)} + NaOH_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(aq)}$

Procedure

Burette solution—base solution

Titration flask—Oxalic acid and HCl respectively

Indicator: Phenolphthalein

(a)Standardization of NaOH solution

A standard solution of oxalic acid is made by measuring a given amount of Oxalic acid (0.63g in 250ml) and dissolving it in water and the volume is made up to 250 ml. Pipette 25ml of the standard oxalic acid solution into a conical

flask and add 2-3 drops of phenolphthalein indicator. Titrate the contents of the flask against NaOH solution obtained from a burette till a permanent light pink color is obtained.

From the results of the titration calculate the concentration of NaOH in moles (molarity), normals (normality) and in g/litre.

b) Determination of the Concentration of the given HCl solution

Pipette 25ml of the given hydrochloric acid into a conical flask and add 2—3 drops of phenolphthalein. Titrate the contents of the flask with NaOH solutions taken in a burette. A light pink color (permanent) appears. Repeat the experiment or titration until two concordant volume values are obtained.

From the results, calculate the concentration of HCl in moles (molarity), normals (normality) and g/liter.

EXPERIMENT 4:

WATER OF HYDRATION

Introduction:

Water has a strong attraction for many compounds because of its polar character and electronic structure. Because water is a normal component of our atmosphere, most compounds will contain some dissolved or absorbed water. In some cases, water molecules become chemically bound in a compound, usually an ionic salt, so that the water molecules become part of the crystal lattice.

In such compounds called hydrates, the water molecules are in e definite proportion relative to the other atoms and must be included as part of the chemical formula.

Copper (II) Sulphate pent hydrate is such a compound. It may be written either as $CuSO_4 (H_2O)_5$ or $CuSO_4.5H_2O$.

The water in hydrates often is loosely bound and may be driven off by heating the solid. If $CuSO_4 (H_2O)_5$ is heated until all the water is driven off, the remaining $CuSO_4$ are called anhydrous Copper (II) Sulphate. If the anhydrous crystals are allowed to stand in the air, they will absorb water from the air continuously, until the pent hydrate is formed. If an aqueous solution of Cu^{2+} and $SO4^{2-}$ ions is evaporated, $CuSO_4 (H_2O)_5$ will crystallize directly. The behavior described for $CuSO_4 (H_2O)_5$ is typical of many other hydrates. Some examples are $NiSO_4 (H_2O)_7$, $NaCO_3 (H_2O)_2$ and $CoCl_2(H_2O)_6$. At any given pressure, the temperature at which a particular hydrate will lose its water completely is difficult for each salt.

Some hydrates lose their water at room temperature and atmospheric pressure; these are called efflorent. Hydrates which are stable at room temperature and one atmosphere, but are not yet saturated with water to stoichiometric limit will absorb water from the atmosphere; these are called hygroscopic. Hygroscopic salts often are used as drying agents, called desiccants.

When a hydrate is heated, it may lose its water in several stages, forming a series of hydrates with regular crystalline structures that contain progressively smaller proportions of water. Color changes often accompany a change in degree of hydration. For example, $CoCl_2 (H_2O)_2$ is violet and anhydrous $CoCl_2$ is blue. Intermediate species such as $CoCl_2 (H_2O)_3$ are unstable and do not form regular crystal structures.

At anytime particular temperature, the degree of stable hydration is well defined, and the stable form of the hydrate has a definite formula.

Plan of experiment

First you examine the behavior of a group of compounds when they are heated, to determine which hydrates are and which are not. Then you drive all the water from unknown hydrate, to determine the mole ration of water in its stable structure.

Safety: Wear approved eye protection

Procedure

A. Testing of hydrates

1. Obtain several different samples from the following group of compounds. Your instructor will tell you how many.

Sodium chloride	Cobalt chloride	Magnesium sulphate
Potassium chloride	Sodium tetra borate	Sodium sulphate
Magnesium chloride Chromium (III) chloride	Sodium acetate	sucrose.
Strontium chloride	Copper (II) sulphate	

   Write the names of your compounds into the table on the data sheet. The degree of hydration for these compounds ranges from zero to twelve water molecules.
2. Place about 0.2g of each compound into separate small dry test tubes. Mark the test tubes for identification.
3. Gently heat each sample, in turn, over a burner flame and observe the results. Write your observations in the table on the data page. Try to infer which of the compounds hydrates are, using the criteria below are.

   Some Characteristics of Hydrate Behavior when Heated.

1. Evolution of water, which may condense on the cooler part wall of the test tube.
2. Color changes may occur in the solid as water is lost.
3. Decomposition may accompany lose of water. The decomposition products often form acids or bases upon reaction with the evolved water of hydration. Test the evolved moisture with litmus or pH paper.

4. After heating, the solid residue often dissolves in water, frequently with a color change.

Questions

1. The formula for a hydrate of sodium phosphate is $Na_3PO_4(H_2O)_{12}$
   (a) If all the water were driven off by heating, how much mass would be lost from a sample which weighed 2.5433g before heating?
   (b) What is the mass % of water in the hydrate?
2. A hydrate of Cobalt chloride, $CoCl_2\ (H_2O)_x$ is heated until it reached a constant mass. The original sample weighed 1.6884g before heating. After heating, its mass was 1.0856g. What is the formula of the hydrate?[use relative atomic mass: Na=23, P=31, H=1, O=16, Co=59, Cl=35.5]

UNIT SCH 2102.

PHYSICAL CHEMISTRY ONE

EXPERIMENT 1:

HESS LAW OF CONSTANT HEAT SUMMATION

The purpose of this experiment is to demonstrate Hess law, which states that the heat change that accompanies a chemical reaction is the same whether it takes place in one or several stages. In this experiment the chemical reaction studied is:

1) $NaOH_{(s)} + H^{+}_{(aq)} + Cl^{-}_{(aq)} \longrightarrow H_2O_{(l)} + Na^{+}_{(aq)} + Cl^{-}_{(aq)}$

Where $H_{(1)}$ is the heat change for the reaction.
This reaction may also take place in two stages:

a) $NaOH_{(s)} \longrightarrow Na^{+}_{(aq)} + OH^{-}_{(aq)}$

b) $Na^{+}_{(aq)} + OH^{-}_{(aq)} + H^{-}_{(aq)} + Cl^{-}_{(aq)} \longrightarrow H_2O_{(l)} + Na^{+}_{(aq)} + Cl^{-}_{(aq)}$

By adding equations (a) and (b), equation 1 is obtained.
It follows from Hess law that when a reaction can be expressed as the algebraic sum of the heats of these other reactions, the heat of the reaction is the algebraic sum of the heats of these other reactions.
Thus,
$H_{(1)} = H(a) + H(b)$
In this experiment H(1), H(a) and H(b) are all determined separately.

Apparatus:
Erlenmeyer flask, 250 cm^3
Beaker, 250 cm^3
Measuring cylinder, 100cm3
Thermometer, 0 to 50 ^0C, graduated to 0.5 ^0C

Reagents:
0.5M Sodium hydroxide
0.5M hydrochloric acid
2g Sodium hydroxide
0.25M hydrochloric acid

Procedure:
Reaction (1):

1. **Weigh a clean dry 250**cm^3 Erlenmeyer flask to 0.1g
2. Pour 200cm^3 of 0.25M HCl into the flask
3. Measure and record the temperature to 0.5^0c
4. Quickly weigh about 2g of dry NaOH pellets to the nearest 0.01g
5. Place the solid NaOH into the flask and swirl to dissolve
6. Record the maximum temperature reached.
7. Record the weights of the solution.

Reaction (a):

1. Wash out the flask used to reaction (1)
2. Repeat the procedure for reaction(1) using 200cm^3 of water in place of 0.25M HCl

Reaction (b):

1. Wash out the flask used for reaction (a).
2. Pour 100cm^3 of 0.5M HCl into the flask.
3. Pour 100cm^3 of 0.5M NaOH into the beaker.
4. When the temperature of both solutions is approximately the same, record the temperature and add the NaOH solution to the HCl solution.
5. Measure and record the maximum temperature reached.
6. Record the weight of the solution.

RESULTS AND CALCULATIONS

Assume the specific heat capacity of glass to be 0.85KJ and that of the solution to be 4.18JK for each reaction.

1. Calculate the change in temperature.
   T=T (f)-T (i)
   Where T (i) and T (f) are the initial and final temperature respectively.
2. Calculate the heat absorbed by the solution
   H(s) =M(s) X T 4.18 J
   Where M(s) = weight of the solution
3. Calculate the heat absorbed by the flask.
   H (f) =M (f) X T X 0.85J
   Where M (f) =weight of the flask
4. Calculate the total heat absorbed.
   H=H(s) + H (f)
5. Calculate the number of moles of NaOH in the reaction(=N)
6. Heat evolved per mole of NaOH
   H=H/N Jmol^{-1}

EXPERIMENT 2

TO FIND THE HEAT OF NEUTRALIZATION OF VARIOUS ACIDS AND BASES

The use of a thermometer calibrated in steps of 0.10C is desirable in this experiment.

Procedure

Pour 50 cm3 of 2M hydrochloric acid from a measuring cylinder into an expanded polystyrene cup, supported in a 250 cm3 beaker. Measure the temperature of the acid, rinse and dry the thermometer and then find the

temperature of 50 cm3 of 2M sodium hydroxide solution in second measuring cylinder. If the two temperatures differ, take an average. Tip the alkali into the acid and gently stir with the thermometer (Vigorous stirring can produce a measurable temperature rise of its own.). Note the maximum temperature reached.
Rinse and dry all the apparatus thoroughly and repeat the experiment using in turn, 50cm3 portion of 2M nitric acid and 1M sulphuric acid in place of the hydrochloric acid. Next, investigate the effect of neutralizing each of the acids with 50cm3 potassium hydroxide solution. Finally repeat the experiment using 50cm3 2M butanoic acid and 50cm3 of 2M ammonia solution.

Questions

1. Why was polystyrene cup used instead of a glass container and why was it placed in the beaker?
2. Work out the enthalpy of neutralization of each acid, i.e. the heat that would be evolved on neutralizing 1 mole of H+ (aq) ions from it. Set out your calculations as shown below;
   You may assume that the density of each solution is 1 gcm3 and that its specific heat capacity is the same as that of water, i.e. $4.2Jg^{-1}k^{-1}$
   No. of moles of $HCl_{(aq)}$ in $50cm^3$ of
   2M hydrochloric acid = 50/100 X 2=0.1
   Total mass of solution=100g
   Initial temperature=T_1
   Final temperature =T_2
   Temperature= (T_2-T_1)
   Heat produced when 0.1 mol of $HCl_{(aq)}$= mass of solution X specific heat capacity/temperature rise.= $100X\ 4.2X\ (T_2-T_1)J$
   Heat produced when 1 mol of $HCl_{(aq)}$=$10\ X\ 100\ X\ 4.2\ X\ (T_2-T_1)J$
   Is neutralized by $NaOH_{(aq)}$
   i.e. Heat of neutralization of hydrochloric acid by sodium hydroxide solution= $10/100\ X\ 100\ X\ 4.2\ X\ (T_2-T_1)\ kJ\ mol^{-1}$

3. Express this result in an equation and in an energy level diagram.

4. Within the limits of experimental error, is there any connection between the results obtained for the strong acids when neutralized by sodium hydroxide solution? Try to explain any pattern, which emerges. (Hint: write an ionic equation for each reaction).

5. Do the results differ noticeably if the acids are neutralized by potassium hydroxide solution? Does this fit in with your explanation in question 4?

6. How does the enthalpy of neutralization of the weak acid, ethanoic acid, with the weak alkali, ammonium solution, compare with that of the others? Can you suggest an explanation for the value obtained in this case?

EXPERIMENT 3:

DETERMINING A SOLUBILITY PRODUCT.

Aim

The purpose of this experiment is to determine the solubility and solubility product of calcium hydroxide

Introduction

The equilibrium between calcium hydroxide and its ions in an aqueous solution is

$Ca(OH)_{2\,(s)} = Ca^{2+}_{(aq)} + 2OH^{-}_{(aq)}$

The concentration of hydroxide ions can be determined by titration with hydrochloric acid; the concentration of calcium ions can be calculated from the titration results.

Requirements

Safety spectacles

4 stoppered bottles, $250cm^3$

Labels for bottles

Spatula

Calcium hydroxide

Measuring cylinder, $100cm^3$

Distilled water

4filter funnels, dry, with filter papers

4 conical flasks, $250cm^3$

Thermometer 0-1000C ($\pm 1^0C$)

Burette and stand, white tile

Pipette, $25cm^3$ and safety filler

Small funnel

Hydrochloric acid solution, 0.1M standardized

Phenolphthalein indicator solution

Procedure

1. Into each of 4 bottles put about 2g of powdered calcium hydroxide and about $100cm^3$ of distilled water. Stopper securely.
2. Shake well for about a minute. Label each bottle with your name.
3. Rinse and fill the burette with standardized HCl

4. Filter the contents of one bottle, allowing the first 5cm^3 to run to and collecting the rest in a dry conical flask. (The first few cm3 are rejected because they are less concentrated in solute than the rest.).The filter paper absorbs solute until it attains equilibrium with the solution. Yet another equilibrium!)
   Step 5 and 6 should be done as quickly as possible (with due care) and with only the minimum shaking that will ensure mixing.
5. Rinse the pipette with the calcium hydroxide solution and transfer 25cm^3 to a conical flask (this need not be dry).
6. Add two drops of phenolphthalein to the flask and titrate the solution until the pink color just disappears. Record your burette readings in a copy of results Table 2.
7. Record the temperature.

Table 2: Results

Solution in flask	mol dm^3		cm^3		
Solution in burette	mol dm^3				
Indicator					
Trial	1	2	3	4	5
Burette Readings Final / Initial					
Volume used					
Mean titre					

Calculation

1. Calculate the concentration of hydroxide ion in a saturated solution of calcium hydroxide.
2. From the equilibrium concentration of hydroxide ion calculate the equilibrium concentration of calcium ion.
3. Calculate the solubility of calcium hydroxide at the temperature of your experiment. Compare your result with the value listed in your data book.
4. Calculate the solubility product from:
   a) Your list
   b) The solubility of $Ca(OH)_2$ given in your data book.
5. Suggest a reason for the speed of working advised for steps 5 and 6 above.
   (Hint: slow working with much shaking of the flask, gives a smaller titre).

EXPERIMENT 4:
A PH OF A WEAK ACID AT VARIOUS CONCENTRATIONS

Aim:

The purpose of this experiment is to examine the effect of dilution on the pH of ethanoic acid, a weak acid.

Introduction:

Ethanoic acid dissociates according to the following equation:

$CH_3CO_2H_{(aq)} = CH_3CO_2^-{}_{(aq)} + H^+_{(aq)}$

The extent of dissociation depends on the initial concentration of acid. By measuring the pH at different concentrations, you can see the effect of solution. These results can be generalized for any weak acid.

Requirements

pH meter with glass electrode
Wash bottle of distilled water
A buffer solution (to calibrate the pH meter)
50cm^3 beaker
0.01M, 0.010M, 0.0010M, 0.00010M ethanoic acid solutions.

Procedure:

1. Calibrate the pH meter by dipping the glass electrode into a solution of known pH(a buffer solution) and turning the adjusting knob so that the scale shows the correct pH value.(If you are in doubt about this ask your teacher)
2. Rinse the glass electrode with distilled water and dip it into a beaker containing 0.00010M ethanoic acid. Record the pH value in a copy of results table 5a. Return the electrode to eater, it must never be dry.
3. Rinse the beaker with the next solution, and repeat step 2, working from the most dilute solution to the most concentrated.
4. Calculate pH values for solutions of HCl at the same concentrations and complete the final column of results table 4a

Results in Table 4a:

Concentration of acid(mol/dm3)	Observed pH of solutions of ethanoic acid	Calculated pH of solutions of hydrochloric acid
0.00010 0.0010 0.010 0.10		

Questions:

1. Compare the pH of ethanoic acid with HCl at each concentration
   a) In which of the two acids is the concentration of hydrogen ions greater?
   b) What does this tells you about the extent of dissociation of ethanoic acid compared to hydrochloric acid?

2.

a) What happens to the difference between the pH of the two acids as concentration decreases?
b) Use Le Chateliers principle to explain the effect of dilution on the extent of dissociation of ethanoic acid.

Wash your hands with soap or detergent and water after handling chromium solutions and the reaction mixtures.

EXPERIMENT 5:

TO DETERMINE HEAT OF NEUTRALIZATION OF STRONG ACID AND STRONG BASE

Introduction:

When aqueous solutions of HCl and NaOH are mixed, a reaction takes place, the products being a salt and water only.

$NaOH_{(aq)} + HCl_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(l)}$

I.e. $OH^- + H^+ \longrightarrow H_2O$

This is a neutralization reaction since the H+ ion which is responsible for acidic properties has reacted with the hydroxide ion which is responsible for basic properties. The purpose of this experiment is to determine the heat change for the neutralization reaction between NaOH and HCl.

Apparatus and Reagents:

2 measuring cylinders, 50ml

1 plastic beaker, 100ml

Thermometer 0—50^{0}C graduated to 0.1 ^{0}C

0.5M HCl

0.5M NaOH

0.5M HNO_3.

Procedure:

Using NaOH as the base and HCl and HNO_3 as the acids, proceed as follows performing the neutralization as duplicate.

1. Pour 40cm^3 of 0.5M NaOH into one measuring cylinder and 40 ml of the acid into the other.
2. Measure accurately and record the temperature of each solution
3. Pour the acid and base together into the beaker simultaneously.

4. Stir gently with the thermometer and record the highest temperature reached.
5. Wash the beaker and do the duplicate.
6. Repeat the above procedure but now use HNO_3 as the acid

Results and Calculations:

1. Calculate the average initial temperature of the acid and base.
2. Calculate the change in temperature $T = T_f - T_{av}$ of the initial
3. Calculate the heat evolved using heat evolved= mass x specific heat capacity X ΔT (for dilute solutions specific heat capacity approximately equals to that of H2O 4.2 J g^{-1} k^{-1}).
4. Calculate the number of moles of water produced.
5. Calculate the heat evolved per mole of water.
6. ΔH (neutralization) = amount of heat evolved.
7. Comment on the ΔH values determined in both cases.

EXPERIMENT 6:

RATE OF THE REACTION BETWEEN- SODIUM THIOSULPHATE AND HYDROCHLORIC ACID

Introduction:

This experiment is designed to examine the kinetics of the reaction between sodium thiosulphate and hydrochloric acid.

$$S_2O_{3\,(aq)} + 2H^+_{(aq)} \longrightarrow S_{(s)} + SO_{2\,(g)} + H_2O_{(l)}$$

This rate is determined from the time it takes a fixed amount of sulpur to precipitate. By varying the concentrations of thiosulphate and then acid, it is possible to determine the order of the reaction with respect to thiosulphate and acid respectively.

Apparatus:

Graduated cylinder, 10cm^3, 100cm^3
Beaker, 50cm^3. Stop watch.

Reagents:

3M hydrochloric acid.
0.15M sodium thiosulphate.

Procedure:

1. Place the beaker on a piece of white paper marked with a cross by a ball point pen. The beaker should be on top of the mark.
2. Add the thiosulphate solution to the beaker to check that the mark is clearly visible through the solution looking from above.
3. Add the acid and time the reaction from the time of addition to the time the mark is completely obscured. Vary the concentration of the thiosulphate as the table below.
4. Repeat the experiment keeping the concentration of thiosulphate constant but varying the acid concentration as in the table below:

cm^3 of 0.15M $NaSO_3$	cm^3 of H_2O	cm^3 of 3M HCl solution
25	0	4
20	5	4
15	10	4
10	15	4
5	20	4
10	0	5
10	1	4

10	2	3
10	3	2
10	4	1

Results and Calculations:

1. Calculate the rate (1/time) for the acid and thiosulphate concentration.
2. Plot:
   i) Change in concentration for both acid and thiosulphate with time.
   ii) Rates of acid and thiosulphate and respective concentration
3. Determine the order of reaction with respect to:
   i) Thiosulphate.
   ii) Acid.

EXPERIMENT 7:

To Compare the Enthalpies of solution of a salt in its anhydrous and hydrated states.

Procedure

Set up the same apparatus as in the previous experiment, pour $50cm^3$ of distilled water into the cup and measure its temperature.

Accurately weigh about 1.6g of anhydrous copper (II) sulphate on a watch glass, tip the solid into the water and stir gently with the thermometer until dissolution is complete. Note the maximum temperature change. (Again, the use of a thermometer calibrated in steps of 0.1 ^{0}C id desirable)
Repeat the experiment, using about 2.5g of copper (II) sulphate in place of the anhydrous salt.

Questions

1. What are the possible sources of error in this experiment?
2. Calculate the enthalpies of solution of the two forms of the salt as shown below. Assume that the densities and the specific heat capacities of the solutions are the same as those of water and that the salt and water were both at the same temperature at the beginning of the experiment.

Mass of copper (II) sulphate =M

Temperature change =T

Heat produced by mass of copper (II) =Mass of solution X specific heat capacity X temp rise

=50 X 4.2 X T J

Heat produced by 1 mol (159.5g) of Copper (II) sulphate dissolves in water $= \frac{159.5 \text{ X } 50 \text{ X } 4.2 \text{ X T J}}{\text{M}}$

i.e.

enthalpy of solution of copper (II) sulphate $= \frac{159.5 \text{ X } 50 \text{ X } 4.2 \text{ T KJ mol}^{-1}}{\text{M X } 1000}$

3. Express your results in the form of energy level diagrams.
4. When solids dissolve, the lattice breaks up and the hydrated ions diffuse throughout the solution. Can you explain the difference in the values of enthalpy of solution obtained for the two forms of the salt?
   (Remember that hydration energy is evolved when ions become hydrated and that energy must be supplied to break up ma crystal lattice.)

EXPERIMENT 8:

CONDUCTANCE OF STRONG AND WEAK ELECTROLYTES.

Plot an equivalent conductance, $\triangle$, against square root of concentration, $\checkmark C$, for solutions of strong electrolytes, is found to be linear at low concentrations. Extrapolation to zero concentration (or infinite dilution) gives the equivalent conductance at zero concentration and is represented by

$\triangle_O$ or (Δ_{oo})

Weak electrolytes, such as acetic acid, are characterized by a rapid non-linear fall of Δ with increase in √C. This rapid decrease is due to a reduction in dissociation of the molecules, whose bonding is predominantly covalent: The degree of dissociation decreases approximately with the square root of concentration. The slight fall of Δ with increase of √C for strong electrolytes is due to long range electrostatic interaction of ions which reduces their ability with increase of concentration.

Materials required

Conductance bridge and dip-type cell, 250ml 0.1 M solutions of potassium chloride and acetic acid, 5.0, 10.0, 25.0 ml pipettes, 100ml graduated flask.

Procedure

Evaluate the cell constant from the measured resistance of the cell when dipping in 0.01M potassium chloride solution.

Prepare 0.05, 0.05, 0.025, 0.01, 0.005, 0.0025, 0.001, 0.0005, and 0.0001 M solutions of potassium chloride by accurate dilutions from the 0.1M solution of potassium chloride using appropriate pipettes and graduated flask. Good quality distilled or de-ionized water of specific conductance less than 1.5×10^{-6} ohm^{-1} should be used for this dilutions. Determine the resistance of the cell when dipping into each of the prepared solutions. Wash out the beaker and cell at least twice with each solution, before taking a measurement. Repeat the dilutions and resistance measurements using acetic acid. If possible, all measurements should be made on the same day to minimize the effects of changes in room temperature. The results obtained using acetic acid solutions also provide the data required for experiment 7.3.

Results.

Tabulate concentration, resistance and the calculated values of conductance, specific conductance, equivalent conductance (for a worked example, see page 90) for each solution investigated.

Plot Δ against √C for potassium chloride and acetic acid. Draw the best straight line through the points for potassium chloride and the best smooth curve through

the points for acetic acid. The results for potassium chloride will probably show a scatter of about 10 percent from a straight line. With careful research procedures, accuracies of about 0.1 percent have been obtained illustrated by the data to be used in experiment 7.2. The errors in the measurements for acetic acid do not show up on the graph to the same extent as those with potassium chloride, because the measured quantity is rapidly reducing in value with increase in concentration.

UNIT SCH 2103

ORGANIC CHEMISTRY ONE

EXPERIMENT 1:

REACTIONS OF HYDROCARBONS

Introduction:

The limited chemical reactivity of the alkanes is indicated by the alternative name, 'paraffins' which means 'little affinity'.

On the other hand, due to the presence of carbon–carbon double bond, alkenes are quite reactive.

In this experiment you will examine some simple reactions of alkanes and alkenes.

EXPERIMENTS:

1. Take three dry test tubes. To one, put 5 drops of an alkane, to the second put 5 drops of an alkene and to the other, 5 drops of benzene.
   – To each of the three test tubes, add ten drops of alkaline potassium permanganate solution.
   – Tabulate your results. [1% $KMnO_{4\,(aq)}$ is made alkaline by the addition of 0.3-0.5 $K_2CO_{3(s)}$].

2. Repeat experiment 1 using 10 drops of acidified potassium permanganate instead of alkaline KMnO4 solution [H_2SO_4 is used to acidify].

3. To 1ml of an alkane in a test tube, add 10- 15 drops of bromine if tetra-chloromethane, drop by drop. Shake the test tube to ensure thorough mixing after addition of each drop. Note any changes in color.

– Repeat using the alkene

– Repeat using the benzene.

4. To each of two 1ml portions of alkane in separate test tubes, add 10–15 drops of bromine in carbon tetra chloride (tetra chloromethane, CCL_4).

After shaking the tubes, place one in the dark (the locker) and expose the other to sunlight for a few minutes.

– Compare the color of the two test tubes. Test for the presence of HBr by holding a moist blue litmus paper above the top of the test tube.

– Finally add 1 ml of distilled water to each test tube, shake and separate the aqueous layer using teat pipettes. Test each aqueous layer with aqueous silver nitrate.

Questions:

1. On the basis of the above experiments, how would you distinguish between cyclopentane and cyclopentene?
2. What kind of reaction occurs when:
   i) An alkene decolorizes bromine in tetra chloromethane.
   ii) An alkene decolorizes bromine in tetra chloromethane in the presence of sunlight?
3. Indicate whether the following statements are true or false.
   i) There is free rotation about the carbo – carbon double bond of the alkenes.
   ii) Alkenes are isometric with cycloalkanes.
   iii) The melting points of unsaturated fats and oils are increased by catalytic hydrogenation.
   iv) The hybridization of the carbon atoms in ethane is changed by an addition reaction.

NB: Give equations in all the experiments where reactions occur.

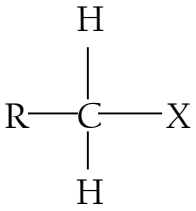
EXPERIMENT 2:

REACTIONS OF ALKYL HALIDES

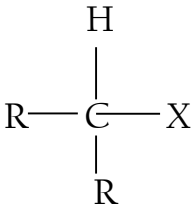
Introduction:

Alkyl halides have the general formula R-X (X=halogen). The halogen is the functional group.

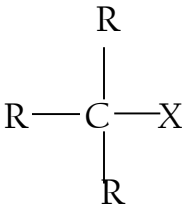
Alkyl halides can be divided into three classes, according to how many alkyl groups are attached to the carbon atom which is bonded to the halogen.



Primary (1^0)



Secondary (2^0)



Tertiary (3^0)

Most important reactions of alkyl halides are those in which the halogen atom, X, is replaced by another group.

$$R{-}X \longrightarrow R{-}OH + NaX$$

A Carbonation

Followed by

$$R^+ + OH^- \longrightarrow R{-}OH$$

(The double barbed curly arrow represents movement of two electrons)

The presence of the X^- ion can be detected by reaction of silver nitrate solution to form insoluble silver halide.

Not all alkyl halides form carbocations easily. The ease of formation depends upon the structure of the halide. The following are some classes of alkyl halides with indication of the ease with which they form carbocation.

Type of halide	Ease of formation of carbocation

Primary;	Difficult
E.g. CH_3CH-Cl	CH_3+CH_2
Secondary;	Easier than primary
E.g. $CH_3\underset{\underset{Cl}{|}}{CH}-CH_3$	$CH_3^{-}+CH-CH_3$
Tertiary;	Easy
E.g. $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$	$CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$
Alkyl;	Easy
E.g. $CH_2=CH-CH_2Cl$	$CH_2=CH-CH_2$

EXPERIMENT:

1. **To compare the rate of hydrolysis of chloro-bromo- and iodoalkanes.**

   -Place 2 ml of ethanol in each of the three test tubes, to act as a solvent. Add to the solvent 5 drops of 1-chloro butane, to the second 5 drops of 1-bromo butane and to the third, 5 drops of 1-iodo butane.

   — Put 5 ml of silver nitrate solution 0.1M in another test tube and stand all four in a beaker of water at about 60^0C for about 10 minutes.

   — Transfer 1 ml of the silver nitrate solution to each of the other three test tubes, shake them to mix their contents and put them back into the warm water. Observe

   a) The order in which the precipitate appears

   b) The color and density of the precipitate

   — Record your observations in tabular form headed experiment, observation and conclusion.

2. To compare the rate of hydrolysis of 1°, 2°, 3° alkyl halides

- Place 2mls of ethanol in each of the three test tubes to act as solvent.
- To the first add 5 drops of a primary alkyl halide, to the second 5 drops of secondary alkyl halide and to the third 5 drops of a tertiary alkyl halide.
- Put 5mls of silver nitrate solution in another test tube and stand all four in a beaker of water at about 60^0C for about 10 minutes.
- Transfer 1 ml of the silver nitrate solution to each of the other three test tubes with shaking and quickly put them back into the warm water.
- Observe the test tubes carefully for about 5 minutes. Alkyl halides which easily give carbocations should form a precipitate within about 5 minutes.
- If no reaction, boil the solution for a further 5 minutes. Secondary alkyl halides usually form a precipitate of silver halide under these conditions but primary halides don't.
- Record your results in tabular form as in experiment 1.
- In three test tubes each containing 2 drops of silver nitrate solution, put 5 drops of sodium chloride to the first one, 5 drops of sodium iodide solution to the second one and 5 drops of sodium bromide to the third one.
- Note down your observations. Is it possible to distinguish between the three halides?

NB: Write equations for all reactions (if any).

Questions:

1. What is the effect on the rate of hydrolysis of alkyl halide arising from changing halogen on the alkyl halides?
2. Give the structural formulae and names of the isomers of C_3H_7Cl. How would you distinguish them chemically? Give equations.

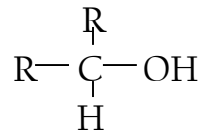
EXPERIMENT 3:

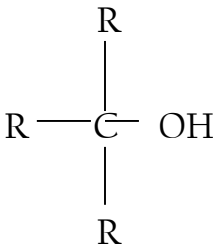
REACTIONS OF ALCOHOLS

Introduction:

Alcohols, also called alkanols, are compounds which contain carbon and hydrogen and a hydrogen and a hydroxyl (OH) group. The OH is the functional group.

Alcohols may be classified into primary $R—CH_2—OH$,

Secondary 

Tertiary 

Reactions of alcohols can be of two main types;

1. Reactions where the oxygen is lost
2. Reactions where is oxygen is retained

Examples in which oxygen is lost:

a) <u>Substitution</u>

$CH_3CH_2OH + X^- \longrightarrow CH_3CH_2—X + OH$

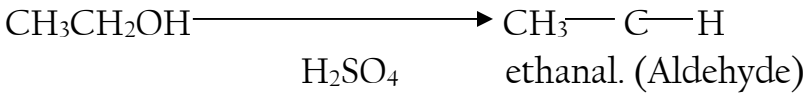
In Lucas test $X = Cl^-$

b) <u>Elimination</u>

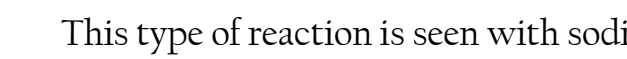
$CH_3CH_2OH \longrightarrow CH_2{=}CH_2{=}CH_2 + H_2O$

Examples in which oxygen is retained: Primary and secondary alcohols can be oxidized by several methods to give aldehydes and ketons respectively. (The aldehydes may be further oxidized to carboxylic acids under some conditions).

$Na_2Cr_7O_7$ O



c) Under strong conditions alcohols can be forced to act as acids



This type of reaction is seen with sodium metal

EXPERIMENT:

1. **Reaction with Sodium**

a) Put about 1 ml of the alcohol in a dry test tube and add a small piece of sodium (rice grain size). Note the effervescence and test the gas with a lighted splint.

b) When the sodium has all dissolved, carefully evaporate the solution to dryness and add 3 drops of water. Test the solution with litmus paper.

2. **Reaction with carboxylic acid (Esterification)**

Warm a mixture of 5 drops of an alcohol and 5 drops of an ethanoic acid with one drop of concentrated sulphuric acid (CARE). Note the characteristic smell of the product.

3. To 5 drops of an alcohol in a test tube add 2-3 drops of ethanoyl chloride.

4. **Oxidation reactions**

a) Place 5 drops of the alcohol in a test tube, add 10 drops of dilute sulphuric acid and 2 drops of potassium dichromate (0.1%).

— Warm gently noting i) the color of the solution ii) the smell of the product

b) Iodoform test: To 5 drops of alcohol add 5 drops of iodine solution (1% solution in 20% solution of potassium iodide) and then dilute sodium hydroxide solution drop wise until the color of the iodine is discharged.

— Note down your observations.

5. The Lucas Test

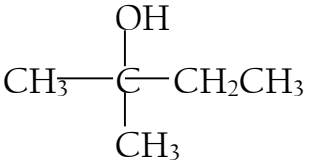
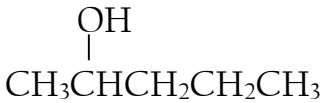
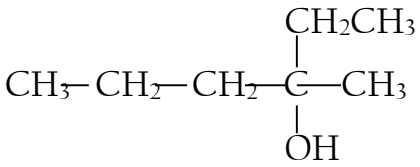
In this test, the OH group of alcohol is substituted by a group. Different classes of alcohol react at different rates.

$R—OH \longrightarrow R—Cl$

(Lucas reagent is made by dissolving 34g of fused zinc chloride in 23g of concentrated hydrochloric acid).

Put 5 drops of the alcohol in a test tube and add 1 ml of the Lucas reagent. Shake the tube and allow to stand for at least 5 minutes. Note down how long the changes occur, if any.

Questions:

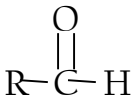
1. Give the IUPAC name for each of the following alcohols
   a) $CH_3CH_2CH_2CH_2OH$
   b)
   
   c)
   
   d)
   
2. Name the four alcohols represented by the molecular formula C_4H_9OH and write their structural formula.
   — What is the effect of oxidation upon each of these compounds?

EXPERIMENT 4:

REACTIONS OF ALDEHYDES AND KETONES

Introduction:

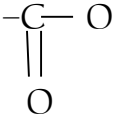
Both aldehydes and ketons contain the carbonyl group(C=O)
In aldehydes, the carbonyl is at the end of a chain and has general

Formula 

Where  is the functional group.

Ketons, have the carbonyl at a none- terminal position of the chain.

Ketons have the general formula 

And the functional group is 

<u>The main reactions of aldehydes and ketons are:</u>

1. Neucleophilic attack on the carbonyl leading to addition. This is often followed by elimination.
2. Removal of the slightly acidic—hydrogen leading to the formation of a carbon ion or enol. This can be followed by condensation of addition.
3. Oxidation—reduction reactions leading to carboxylic acids or alcohols respectively.

   Perform the following series of experiments to investigate the reactions of aldehydes and ketons.

1. <u>The Tollen's reagent(Silver mirror test)</u>

This test makes use of the reduction of silver (I) ion to metallic silver.

— put about 1 ml of 5% silver nitrate solution in a clean test tube, add 3 4 drops of sodium hydroxide(2 M)followed by drop wise addition of ammonium hydroxide(0.1M) until the precipitate has just redissolved.

— Then add 3–4 drops of the aldehyde or ketone and warm gently on water bath shaking for about 5 minutes. Note down your observations.

2. Fehling's solution

— To 1ml of the aldehyde or ketone, add 1ml of 10% sodium carbonate solution followed by 1ml of Fehling's solution and boil the mixture for 1 minute. Note down your observation.

3. The Schiff's Fuschin test

This is a very sensitive test. The reagent should **never be warmed** or a false result may be obtained.

— Add 1ml of Schiff's reagent to 1ml of the aldehyde or ketone. Note down your observation. If an aldehyde, a deep purple color should result.

4. Reaction with potassium dichromate

— To about 2mls of a ketone or aldehyde, add 3ml of potassium dichromate(1%) and a few drops of concentrated sulphuric acid **(Care: corrosive)**. Warm the mixture gently.

5. Reaction with potassium permanganate solution

— To about 2mls of an aldehyde or ketone, add about 3ml of 1% potassium permanganate solution and a few drops of concentrated sulphuric acid. Warm the mixture gently.

6. The iodoform test

— the reagent in iodoform test reacts with aldehydes and ketones that have the structural feature $CH_3—C=O$

For an aldehyde R=H and hence only ethanol $CH_3—C=O$ gives positive results.
Alcohol which on oxidation gives $R—C=CH_3$ also give positive iodoform reaction.

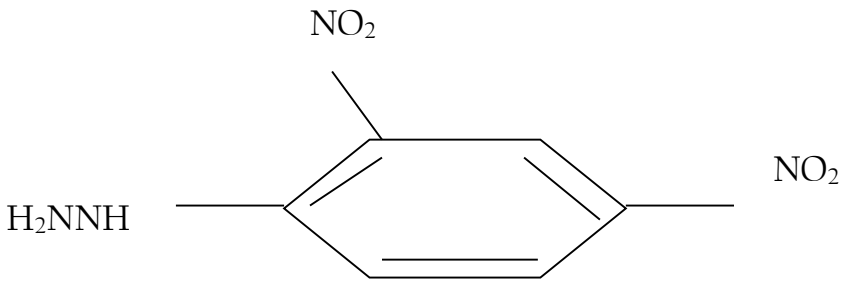
- Add 3–4 drops of the aldehyde or ketone to 2ml of water and add 2ml of 10% sodium hydroxide solution.
- Add drop wise with shaking a 10% solution of iodine in 20% potassium iodide solution until a dark brown color persists.
- Allow to stand and warm if necessary.

A positive result is indicated by the presence of yellow crystals.

- Record down your observations.

7. Addition-Elimination(Condensation)reactions with 2,4- Dinitrophenylhydrazine

Most aldehydes and ketones react with
2, 4—Dinitrophenylhydrazine (2, 4—DNPH) whose structure is



The reaction involves the addition to the carbonyl group followed by elimination.

- Take a few drops of aldehyde or ketone and dissolve in the minimum of methanol.
- Add to this solution about 5ml of 2, 4—DNPH reagent and allow to stand. If no precipitation, add 1 —2ml of dilute sulphuric acid.
- Record down your observations.

Summarize your results from reactions 1 to 7 on aldehydes and ketones in one or two paragraphs.